

LINUX COMMANDS

Chapter 1:

1. #uname

```
Cloud Shell Editor

cloudshell x + v

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

arunsubramaniyan323@cloudshell:~$ uname
Linux
arunsubramaniyan323@cloudshell:~$
```

2. #uptime

```
arunsubramaniyan323@cloudshell:~$ uptime
10:14:06 up 42 min, 0 user, load average: 0.00, 0.06, 0.04
arunsubramaniyan323@cloudshell:~$
```

3. #hostname

```
arunsubramaniyan323@cloudshell:~$ hostname
cs-377686329843-default
arunsubramaniyan323@cloudshell:~$
arunsubramaniyan323@cloudshell:~$
```

4. #dmesg

```
arunsubramaniyan323@cloudshell:~$ dmesg
dmesg: read kernel buffer failed: Operation not permitted
arunsubramaniyan323@cloudshell:~$
```

5. #free

```

arunsubramaniyan323@cloudshell:~$ free
              total        used        free      shared  buff/cache   available
Mem:      16382844       911460      14383664         1148     1366564     15471384
Swap:            0              0              0
arunsubramaniyan323@cloudshell:~$

```

6. #top

```

top - 10:19:27 up 48 min, 0 user, load average: 0.00, 0.04, 0.03
Tasks: 30 total, 2 running, 28 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.7 us, 1.1 sy, 0.0 ni, 97.1 id, 0.0 wa, 0.0 hi, 0.1 si, 0.1 st
MiB Mem : 15998.9 total, 14043.6 free, 893.0 used, 1334.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 15105.9 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
  451 root        20   0 1867212  46484 32552 S   0.3   0.3   0:01.34 containerd
    1 root        20   0   4324     3264  2984 S   0.0   0.0   0:00.11 bash
    9 syslog      20   0 222140     3888  2536 S   0.0   0.0   0:00.02 rsyslogd
   25 root        20   0   38768  27988 10888 S   0.0   0.2   0:00.82 python
   26 root        20   0   6196     2684  2440 S   0.0   0.0   0:00.00 logger
   67 root        10 -10  12024     4176  3064 S   0.0   0.0   0:00.01 sshd
  302 root        20   0 2169572  84832 60800 S   0.0   0.5   0:00.71 dockerd
  436 root        20   0 1234524  12616  8248 S   0.0   0.1   0:00.44 editor-proxy
  437 root        20   0   16328   6424  5500 S   0.0   0.0   0:00.02 sudo
  442 root        20   0 1226356   2852  1976 S   0.0   0.0   0:00.01 tmux-agent
  467 root        10 -10  13996   9724  8348 S   0.0   0.1   0:00.03 sshd
  474 root        10 -10  13996   9824  8452 S   0.0   0.1   0:00.03 sshd
  476 root        20   0   9196   4156  3780 S   0.0   0.0   0:00.00 runuser
  477 arunsub+    20   0   4324   3336  3068 S   0.0   0.0   0:00.01 sh
  499 arunsub+    20   0  11.3g 155084 50820 R   0.0   0.9   0:06.19 node
  522 arunsub+    10 -10  14256   6360  4704 S   0.0   0.0   0:00.00 sshd
  589 arunsub+    10 -10  14256   6476  4820 S   0.0   0.0   0:00.04 sshd
  599 arunsub+    10 -10   7340   3824  3576 S   0.0   0.0   0:00.00 bash
  604 arunsub+    10 -10   7604   4480  3868 S   0.0   0.0   0:00.00 bash
  659 arunsub+    10 -10   7340   3756  3504 S   0.0   0.0   0:00.00 bash
  662 arunsub+    10 -10   7340   3548  3284 S   0.0   0.0   0:00.00 start-shell.sh
  664 arunsub+    10 -10   6880   3284  2984 S   0.0   0.0   0:00.00 tmux
  670 arunsub+    10 -10   7280   2804  2028 S   0.0   0.0   0:00.05 tmux
  671 arunsub+    10 -10   8800   5812  3948 S   0.0   0.0   0:00.11 bash
  753 root        20   0   2696   1572  1476 S   0.0   0.0   0:00.00 sleep
 1492 arunsub+    20   0  11.3g 96532 50488 S   0.0   0.6   0:01.71 node
 1521 arunsub+    20   0 1048772  61392 47572 S   0.0   0.4   0:00.32 node
 1570 arunsub+    20   0 1062156  75620 48024 S   0.0   0.5   0:00.86 node
 1593 arunsub+    20   0   8920   6124  4116 S   0.0   0.0   0:00.03 bash
 1971 arunsub+    10 -10  11924   5956  3744 R   0.0   0.0   0:00.00 top

```

7. #htop

```

0 [ ] 2.7s Tasks: 30, 74 thr, 0 kthr, 2 running
1 [ ] 1.3s Load average: 0.10 0.06 0.03
2 [ ] 2.0s Uptime: 00:48:51
3 [ ] 2.0s
Mem [ ] 625M/15.4G
Swp [ ] 0K/0K

Main I/O
PID USER      PRI  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  Command
463 root        20   0 1823M 46484 32552 S   0.0  0.3   0:00.36 containerd --config /var/run/docker/containers/containerd.toml
  1 root        20   0   4324     3266  2984 S   0.0  0.0   0:00.09 /bin/bash /google/scripts/onrun.sh sleep infinity
  9 syslog      20   0 216M 3886 2536 S   0.0  0.0   0:00.00 /usr/sbin/rsyslogd
11 syslog      20   0 216M 3886 2536 S   0.0  0.0   0:00.00 /usr/sbin/rsyslogd
12 syslog      20   0 216M 3886 2536 S   0.0  0.0   0:00.00 /usr/sbin/rsyslogd
25 root        20   0 38768 27988 10888 S   0.0  0.2   0:00.83 /usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --logfile
26 root        20   0   6196     2684  2440 S   0.0  0.0   0:00.00 logger -t supervisord
67 root        10 -10  12024     4176  3064 S   0.0  0.0   0:00.01 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/keys/authorized_keys [listener] 0 of 10-60 startups
302 root        20   0 2118M 84832 60800 S   0.0  0.5   0:00.31 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
407 root        20   0 2118M 84832 60800 S   0.0  0.5   0:00.08 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
408 root        20   0 2118M 84832 60800 S   0.0  0.5   0:00.01 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
409 root        20   0 2118M 84832 60800 S   0.0  0.5   0:00.07 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
410 root        20   0 2118M 84832 60800 S   0.0  0.5   0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
411 root        20   0 2118M 84832 60800 S   0.0  0.5   0:00.06 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
434 root        20   0 2118M 84832 60800 S   0.0  0.5   0:00.02 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
436 root        20   0 1205M 12616 8248 S   0.0  0.1   0:00.03 /google/devshell/editor/proxy/editor-proxy --static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-c
437 root        20   0 1328 6424 5500 S   0.0  0.0   0:00.02 sudo /google/devshell/tmux-agent
540 arunsubram  20   0 11.3G 151M 50820 S   0.0  0.9   0:00.57 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.j
541 arunsubram  20   0 11.3G 151M 50820 S   0.0  0.9   0:00.54 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.j
542 arunsubram  20   0 11.3G 151M 50820 S   0.0  0.9   0:00.47 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.j
549 root        20   0 2118M 84836 60800 S   0.0  0.5   0:00.02 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
566 arunsubram  20   0 11.3G 151M 50820 S   0.0  0.9   0:00.02 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.j
589 arunsubram  10 -10 14256 6476 4820 S   0.0  0.0   0:00.05 sshd: arunsubramaniyan323@pts/1: /usr/bin/sudo -E --no-profiles 0 /usr/bin/sudo rm -f /var/run/google/devshell/35225" EXIT
599 arunsubram  10 -10 7340 3824 3576 S   0.0  0.0   0:00.00 bash -c (sudo touch /var/run/google/devshell/35225 &) registry-mirror trap 'sudo rm -f /var/run/google/devshell/35225" EXIT
604 arunsubram  10 -10 7604 4480 3868 S   0.0  0.0   0:00.00 /bin/bash --nrc --no-profiles 0 /usr/bin/sudo rm -f /var/run/google/devshell/35225" EXIT
659 arunsubram  10 -10 7340 3756 3504 S   0.0  0.0   0:00.00 bash -c if [ -f /google/devshell/start-shell.sh ]; then /google/devshell/start-shell.sh '' '' '' '631831535' false
662 arunsubram  10 -10 7340 3548 3284 S   0.0  0.0   0:00.00 /bin/bash /google/devshell/start-shell.sh 631831535 false /devshell/editor/code-oss-for-cloud-shell/out/server-main.j

```

8. #lsb_release

```
arunsubramaniyan323@cloudshell:~$ lsb_release
No LSB modules are available.
arunsubramaniyan323@cloudshell:~$
```

9. #arch

```
arunsubramaniyan323@cloudshell:~$ arch
x86_64
arunsubramaniyan323@cloudshell:~$
```

Chapter 2:

```
arunsubramaniyan323@cloudshell:~$ ls
nura  README-cloudshell.txt
arunsubramaniyan323@cloudshell:~$ cd nura
arunsubramaniyan323@cloudshell:~/nura$ pwd
/home/arunsubramaniyan323/nura
arunsubramaniyan323@cloudshell:~/nura$ mkdir linux
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  linux  main
arunsubramaniyan323@cloudshell:~/nura$ touch linux1
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  linux  linux1  main
arunsubramaniyan323@cloudshell:~/nura$ rmdir linux
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  linux1  main
arunsubramaniyan323@cloudshell:~/nura$ rm linux1
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main
arunsubramaniyan323@cloudshell:~/nura$ cp main linux
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  linux  main
arunsubramaniyan323@cloudshell:~/nura$ mv linux demo.py
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main
arunsubramaniyan323@cloudshell:~/nura$ ln demo.py main
ln: failed to create hard link 'main': File exists
arunsubramaniyan323@cloudshell:~/nura$ mv demo.py test.py
arunsubramaniyan323@cloudshell:~/nura$ ls
main  test.py
arunsubramaniyan323@cloudshell:~/nura$ ln test.py demo.py
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$
```

Chapter 3:

```
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$ cat demo.py
print("Hello World")
print("Today we learned LINUX commands:")
print("System information commands and file and directory commands")
arunsubramaniyan323@cloudshell:~/nura$ nano test.py
arunsubramaniyan323@cloudshell:~/nura$ cat test.py
print("Hello World")
print("Today we learned LINUX commands:")
print("System information commands and file and directory commands")
arunsubramaniyan323@cloudshell:~/nura$ nano test.py
arunsubramaniyan323@cloudshell:~/nura$ cat test.py

nums = [1,3,2,4]

def one_bit_diff(a, b):
    x = a ^ b
    return x != 0 and (x & (x - 1)) == 0

def valid_gray(nums):

    if nums[0] != 0:
        return False

    if len(nums) != len(set(nums)):
        return False

    for i in range(len(nums) - 1):
        if not one_bit_diff(nums[i], nums[i + 1]):
            return False

    if not one_bit_diff(nums[0], nums[-1]):
        return False

    return True

print(str(valid_gray(nums)).lower())
arunsubramaniyan323@cloudshell:~/nura$
```

```
arunsubramaniyan323@cloudshell:~/nura$ more test.py
```

```
nums = [1,3,2,4]
```

```
def one_bit_diff(a, b):  
    x = a ^ b  
    return x != 0 and (x & (x - 1)) == 0
```

```
def valid_gray(nums):  
  
    if nums[0] != 0:  
        return False  
  
    if len(nums) != len(set(nums)):  
        return False  
  
    for i in range(len(nums) - 1):  
        if not one_bit_diff(nums[i], nums[i + 1]):  
            return False  
  
    if not one_bit_diff(nums[0], nums[-1]):  
        return False  
  
    return True
```

```
print(str(valid_gray(nums)).lower())
```

```
arunsubramaniyan323@cloudshell:~/nura$
```



```
VIM - Vi IMproved

        version 9.1.697
        by Bram Moolenaar et al.
    Modified by team+vim@tracker.debian.org
    Vim is open source and freely distributable
```

```
        Help poor children in Uganda!
type  :help iccf<Enter>      for information

type  :q<Enter>              to exit
type  :help<Enter> or <F1>   for on-line help
type  :help version9<Enter> for version info
```

```
arunsubramaniyan323@cloudshell:~$ ls
nura README-cloudshell.txt
arunsubramaniyan323@cloudshell:~$ gedit
-bash: gedit: command not found
arunsubramaniyan323@cloudshell:~$
```

```
arunsubramaniyan323@cloudshell:~$ awk
Usage: mawk [Options] [Program] [file ...]
```

Program:

The -f option value is the name of a file containing program text.
If no -f option is given, a "--" ends option processing; the following
parameters are the program text.

Options:

-f program-file Program text is read from file instead of from the
command-line. Multiple -f options are accepted.
-F value sets the field separator, FS, to value.
-v var=value assigns value to program variable var.
-- unambiguous end of options.

Implementation-specific options are prefixed with "-W". They can be
abbreviated:

-W version show version information and exit.
-W compat pre-POSIX 2001.
-W dump show assembler-like listing of program and exit.
-W help show this message and exit.
-W interactive set unbuffered output, line-buffered input.
-W exec file use file as program as well as last option.
-W posix stricter POSIX checking.
-W random=number set initial random seed.
-W sprintf=number adjust size of sprintf buffer.
-W usage show this message and exit.

```
arunsubramaniyan323@cloudshell:~$
```

```

arunsubramaniyan323@cloudshell:~$ sed
Usage: sed [OPTION]... {script-only-if-no-other-script} [input-file]...

  -n, --quiet, --silent
                suppress automatic printing of pattern space
      --debug
                annotate program execution
  -e script, --expression=script
                add the script to the commands to be executed
  -f script-file, --file=script-file
                add the contents of script-file to the commands to be executed
--follow-symlinks
                follow symlinks when processing in place
  -i[SUFFIX], --in-place[=SUFFIX]
                edit files in place (makes backup if SUFFIX supplied)
  -l N, --line-length=N
                specify the desired line-wrap length for the 'l' command
--posix
                disable all GNU extensions.
  -E, -r, --regexp-extended
                use extended regular expressions in the script
                (for portability use POSIX -E).
  -s, --separate
                consider files as separate rather than as a single,
                continuous long stream.
      --sandbox
                operate in sandbox mode (disable e/r/w commands).
  -u, --unbuffered
                load minimal amounts of data from the input files and flush
                the output buffers more often
  -z, --null-data
                separate lines by NUL characters
      --help    display this help and exit
      --version output version information and exit

If no -e, --expression, -f, or --file option is given, then the first
non-option argument is taken as the sed script to interpret.  All
remaining arguments are names of input files; if no input files are
specified, then the standard input is read.

GNU sed home page: <https://www.gnu.org/software/sed/>.
General help using GNU software: <https://www.gnu.org/gethelp/>.
arunsubramaniyan323@cloudshell:~$

```

Chapter 4:

```

arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$ chmod u+x test.py
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$

arunsubramaniyan323@cloudshell:~/nura$ sudo chown arunsubramaniyan323 test.py
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$

```

```
arunsubramaniyan323@cloudshell:~/nura$ sudo chgrp arunsubramaniyan323 test.py
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$

arunsubramaniyan323@cloudshell:~/nura$ umask
0002
arunsubramaniyan323@cloudshell:~/nura$

arunsubramaniyan323@cloudshell:~/nura$ stat test.py
  File: test.py
  Size: 455          Blocks: 8          IO Block: 4096   regular file
Device: 8,17    Inode: 131382        Links: 2
Access: (0764/-rwxrw-r--)  Uid: ( 1000/arunsubramaniyan323)   Gid: ( 1000/arunsubramaniyan323)
Access: 2026-05-14 10:29:40.078075182 +0000
Modify: 2026-05-14 10:39:54.751213195 +0000
Change: 2026-05-14 17:25:46.935574261 +0000
 Birth: 2026-05-14 10:29:40.078075182 +0000
arunsubramaniyan323@cloudshell:~/nura$

arunsubramaniyan323@cloudshell:~/nura$ lsattr test.py
-----e----- test.py
arunsubramaniyan323@cloudshell:~/nura$

arunsubramaniyan323@cloudshell:~/nura$ sudo chattr +i test.py
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$
```

Chapter 5

```
arunsubramaniyan323@cloudshell:~/nura$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 4a:cc:35:70:c0:f9 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.4 netmask 255.255.0.0 broadcast 10.88.255.255
    ether ce:b9:74:e6:0b:0e txqueuelen 1000 (Ethernet)
    RX packets 30136 bytes 14183676 (14.1 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 24962 bytes 38755036 (38.7 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 23086 bytes 82782129 (82.7 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 23086 bytes 82782129 (82.7 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

arunsubramaniyan323@cloudshell:~/nura$ 

arunsubramaniyan323@cloudshell:~/nura$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0@if8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc noqueue state UP group default qlen 1000
    link/ether ce:b9:74:e6:0b:0e brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.88.0.4/16 brd 10.88.255.255 scope global eth0
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1460 qdisc noqueue state DOWN group default
    link/ether 4a:cc:35:70:c0:f9 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
arunsubramaniyan323@cloudshell:~/nura$ 

arunsubramaniyan323@cloudshell:~/nura$ ip route
default via 10.88.0.1 dev eth0
10.88.0.0/16 dev eth0 proto kernel scope link src 10.88.0.4
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
arunsubramaniyan323@cloudshell:~/nura$ 
```

```

arunsubramaniyan323@cloudshell:~/nura$ ping google.com
PING google.com (64.233.170.138) 56(84) bytes of data.
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=1 ttl=114 time=0.895 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=2 ttl=114 time=0.582 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=3 ttl=114 time=0.605 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=4 ttl=114 time=0.552 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=5 ttl=114 time=0.566 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=6 ttl=114 time=0.552 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=7 ttl=114 time=0.735 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=8 ttl=114 time=0.655 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=9 ttl=114 time=0.596 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=10 ttl=114 time=0.624 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=11 ttl=114 time=0.527 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=12 ttl=114 time=0.659 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=13 ttl=114 time=0.664 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=14 ttl=114 time=0.527 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=15 ttl=114 time=0.558 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=16 ttl=114 time=0.611 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=17 ttl=114 time=0.656 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=18 ttl=114 time=0.543 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=19 ttl=114 time=0.559 ms
64 bytes from sg-in-fl38.1e100.net (64.233.170.138): icmp_seq=20 ttl=114 time=0.677 ms
^C
--- google.com ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19448ms
rtt min/avg/max/mdev = 0.527/0.617/0.895/0.084 ms
arunsubramaniyan323@cloudshell:~/nura$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:34549         0.0.0.0:*               LISTEN
tcp        0      0 localhost:8998          0.0.0.0:*               LISTEN
tcp        0      0 localhost:33583         0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:971            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp        0      0 localhost:35579         0.0.0.0:*               LISTEN
tcp        0      0 localhost:54330         localhost:970           ESTABLISHED
tcp        0      0 localhost:44158         localhost:ssh           TIME_WAIT
tcp        0      0 localhost:54074         localhost:ssh           ESTABLISHED
tcp        0      0 localhost:58286         localhost:ssh           ESTABLISHED
tcp        0      0 localhost:ssh           localhost:54074         ESTABLISHED
tcp        0      0 localhost:ssh           localhost:58280         ESTABLISHED
tcp        0      0 localhost:ssh           localhost:58286         ESTABLISHED
tcp        0      0 localhost:37498         localhost:971           ESTABLISHED
tcp        0      0 localhost:44144         localhost:ssh           TIME_WAIT
tcp        0      0 localhost:971           localhost:37490         ESTABLISHED
tcp        0      0 cs-377686329843-d:45966 sa-in-fl101.1e100.:https ESTABLISHED
tcp        0      0 localhost:58020         localhost:ssh           TIME_WAIT
tcp        0      0 localhost:58280         localhost:ssh           ESTABLISHED
tcp        0      0 localhost:971           localhost:37498         ESTABLISHED
tcp        0      0 localhost:54324         localhost:970           ESTABLISHED
tcp        0      0 localhost:37490         localhost:971           ESTABLISHED
tcp6       0      0 [::]:980               [::]:*                 LISTEN
tcp6       0      0 [::]:981                [::]:*                 LISTEN
tcp6       0      0 [::]:970                [::]:*                 LISTEN
tcp6       0      0 [::]:922                [::]:*                 LISTEN
tcp6       0      0 [::]:900                [::]:*                 LISTEN
tcp6       0      0 [::]:ssh                [::]:*                 LISTEN
tcp6       0 1128 cs-377686329843-def:922 180.76.57.208:56136    ESTABLISHED
tcp6       0 574 cs-377686329843-def:980 10.111.157.14:57052    ESTABLISHED
tcp6       0      0 localhost:970           localhost:54324         ESTABLISHED
tcp6       0      0 cs-377686329843-def:980 10.111.157.12:56200    ESTABLISHED
tcp6       0      0 cs-377686329843-def:980 10.111.157.12:47164    ESTABLISHED
tcp6       0      0 cs-377686329843-def:980 10.111.157.14:36412    ESTABLISHED
tcp6       0      0 cs-377686329843-def:980 10.111.157.12:47158    ESTABLISHED

```

```

tcp6      0      0 localhost:970      localhost:54330    ESTABLISHED
tcp6      0      0 cs-377686329843-def:980 10.111.157.12:47172 ESTABLISHED
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags      Type       State      I-Node    Path
unix  3      [ ]       STREAM     CONNECTED  44682
unix  2      [ ACC ]   STREAM     LISTENING  21123     /var/run/docker/metrics.sock
unix  3      [ ]       STREAM     CONNECTED  44423     /tmp/tmux-1000/default
unix  3      [ ]       STREAM     CONNECTED  43948
unix  3      [ ]       STREAM     CONNECTED  21124     /var/run/docker/containerd/containerd.sock
unix  3      [ ]       STREAM     CONNECTED  22496
unix  2      [ ACC ]   STREAM     LISTENING  21298     /var/run/docker/libnetwork/313fd1c89ff7.sock
unix  3      [ ]       STREAM     CONNECTED  44396
unix  2      [ ]       DGRAM      CONNECTED  21722
unix  3      [ ]       STREAM     CONNECTED  43954
unix  3      [ ]       STREAM     CONNECTED  46446
unix  2      [ ACC ]   STREAM     LISTENING  22047     /tmp/tmux-1000/default
unix  3      [ ]       STREAM     CONNECTED  44679
unix  2      [ ]       DGRAM      CONNECTED  21867
unix  3      [ ]       STREAM     CONNECTED  43951
unix  3      [ ]       STREAM     CONNECTED  43932
unix  2      [ ]       STREAM     CONNECTED  46336
unix  2      [ ACC ]   STREAM     LISTENING  22499     /tmp/vscode-ipc-858c7182-a0e4-428d-a66d-9123769c0e42.sock
unix  3      [ ]       STREAM     CONNECTED  44721
unix  3      [ ]       STREAM     CONNECTED  46445
unix  3      [ ]       STREAM     CONNECTED  44414
unix  2      [ ]       DGRAM      CONNECTED  21076
unix  2      [ ACC ]   STREAM     LISTENING  44560     /tmp/vscode-ipc-9c53751a-140d-4dfd-alb2-3190de3f6d13.sock
unix  2      [ ACC ]   STREAM     LISTENING  44660     /tmp/vscode-git-890529dc5e.sock
unix  3      [ ]       STREAM     CONNECTED  43949
unix  2      [ ACC ]   STREAM     LISTENING  23745     /var/run/docker.sock
unix  3      [ ]       STREAM     CONNECTED  44720
unix  3      [ ]       STREAM     CONNECTED  44685
unix  3      [ ]       STREAM     CONNECTED  46348
unix  3      [ ]       STREAM     CONNECTED  43952
unix  3      [ ]       STREAM     CONNECTED  54549
unix  3      [ ]       STREAM     CONNECTED  44718
unix  3      [ ]       STREAM     CONNECTED  22497
unix  2      [ ]       DGRAM      CONNECTED  23141
unix  3      [ ]       STREAM     CONNECTED  43930
unix  3      [ ]       STREAM     CONNECTED  44686
unix  3      [ ]       STREAM     CONNECTED  23807
unix  3      [ ]       STREAM     CONNECTED  43955
unix  3      [ ]       STREAM     CONNECTED  44684
unix  3      [ ]       STREAM     CONNECTED  46442
unix  3      [ ]       STREAM     CONNECTED  46444
unix  3      [ ]       STREAM     CONNECTED  44415
unix  3      [ ]       STREAM     CONNECTED  44681
unix  9      [ ]       DGRAM      CONNECTED  22565     /dev/log
unix  3      [ ]       STREAM     CONNECTED  44722
unix  3      [ ]       STREAM     CONNECTED  46443
unix  3      [ ]       STREAM     CONNECTED  43953
unix  2      [ ]       DGRAM      CONNECTED  44409
unix  3      [ ]       STREAM     CONNECTED  45440
unix  3      [ ]       STREAM     CONNECTED  45439
unix  3      [ ]       STREAM     CONNECTED  54548
unix  3      [ ]       STREAM     CONNECTED  44683
unix  3      [ ]       STREAM     CONNECTED  46441
unix  3      [ ]       STREAM     CONNECTED  43950
unix  3      [ ]       STREAM     CONNECTED  43929
unix  3      [ ]       STREAM     CONNECTED  22845     /var/run/docker/containerd/containerd.sock
unix  3      [ ]       STREAM     CONNECTED  46447
unix  2      [ ]       DGRAM      CONNECTED  44390
unix  3      [ ]       STREAM     CONNECTED  44717
unix  2      [ ]       DGRAM      CONNECTED  22778
unix  3      [ ]       STREAM     CONNECTED  43928
unix  3      [ ]       STREAM     CONNECTED  21903
unix  3      [ ]       STREAM     CONNECTED  44680
unix  3      [ ]       STREAM     CONNECTED  43927
unix  3      [ ]       STREAM     CONNECTED  44719
unix  2      [ ]       STREAM     CONNECTED  44379
unix  2      [ ACC ]   STREAM     LISTENING  23776     /var/run/docker/containerd/containerd-debug.sock
unix  2      [ ACC ]   STREAM     LISTENING  23777     /var/run/docker/containerd/containerd.sock.ttrpc
unix  3      [ ]       STREAM     CONNECTED  43931
unix  3      [ ]       STREAM     CONNECTED  46440
unix  2      [ ACC ]   STREAM     LISTENING  23778     /var/run/docker/containerd/containerd.sock

```

```

arunsubramaniyan323@cloudshell:~/nura$ █

```

```
arunsubramaniyan323@cloudshell:~/nura$ ss -tln
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port      Process
tcp        LISTEN     0           511         127.0.0.1:34549          0.0.0.0:*              *
tcp        LISTEN     0           5           127.0.0.1:8998           0.0.0.0:*              *
tcp        LISTEN     0           511         127.0.0.1:33583          0.0.0.0:*              *
tcp        LISTEN     0           511         0.0.0.0:971             0.0.0.0:*              *
tcp        LISTEN     0           128         0.0.0.0:22              0.0.0.0:*              *
tcp        LISTEN     0           128         127.0.0.1:35579          0.0.0.0:*              *
tcp        LISTEN     0           4096        *:980                   *:~                      *
tcp        LISTEN     0           4096        *:981                   *:~                      *
tcp        LISTEN     0           4096        *:970                   *:~                      *
tcp        LISTEN     0           4096        *:922                   *:~                      *
tcp        LISTEN     0           4096        *:900                   *:~                      *
tcp        LISTEN     0           128         [::]:22                 [::]:~                  *
```

```
arunsubramaniyan323@cloudshell:~/nura$ sudo apt install traceroute
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  traceroute
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 60.5 kB of archives.
After this operation, 162 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 traceroute amd64 1:2.1.5-1 [60.5 kB]
Fetched 60.5 kB in 0s (1,570 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package traceroute.
(Reading database ... 121190 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.5-1_amd64.deb ...
Unpacking traceroute (1:2.1.5-1) ...
Setting up traceroute (1:2.1.5-1) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute.1.gz because associated file /usr/share/man/man1/traceroute.db.1.gz (of link group traceroute) doesn't exist
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceroute6.1.gz because associated file /usr/share/man/man1/traceroute6.db.1.gz (of link group traceroute6) doesn't exist
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/lft.1.gz because associated file /usr/share/man/man1/lft.db.1.gz (of link group lft) doesn't exist
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/traceproto.1.gz because associated file /usr/share/man/man1/traceproto.db.1.gz (of link group traceproto) doesn't exist
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man8/tcptraceroute.8.gz because associated file /usr/share/man/man8/tcptraceroute.db.8.gz (of link group tcptraceroute) doesn't exist
Processing triggers for man-db (2.12.0-4build2) ...
arunsubramaniyan323@cloudshell:~/nura$
```

```
arunsubramaniyan323@cloudshell:~/nura$ wget https://example.com/file.zip
--2026-05-14 17:38:21-- https://example.com/file.zip
Resolving example.com (example.com)... 172.66.147.243, 104.20.23.154, 2606:4700:10::6814:179a, ...
Connecting to example.com (example.com)|172.66.147.243|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-05-14 17:38:21 ERROR 404: Not Found.
```

```
arunsubramaniyan323@cloudshell:~/nura$
```

```
arunsubramaniyan323@cloudshell:~/nura$ curl https://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}hl{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>
```

```

arunsubramaniyan323@cloudshell:~/nura$ nslookup google.com
Server:      169.254.169.254
Address:     169.254.169.254#53

Non-authoritative answer:
Name:   google.com
Address: 64.233.170.102
Name:   google.com
Address: 64.233.170.138
Name:   google.com
Address: 64.233.170.139
Name:   google.com
Address: 64.233.170.113
Name:   google.com
Address: 64.233.170.101
Name:   google.com
Address: 64.233.170.100
Name:   google.com
Address: 2404:6800:4003:c00::8a
Name:   google.com
Address: 2404:6800:4003:c00::66
Name:   google.com
Address: 2404:6800:4003:c00::64
Name:   google.com
Address: 2404:6800:4003:c00::8b

arunsubramaniyan323@cloudshell:~/nura$ 
arunsubramaniyan323@cloudshell:~/nura$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 62114
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                281     IN      A      64.233.170.102
google.com.                281     IN      A      64.233.170.138
google.com.                281     IN      A      64.233.170.139
google.com.                281     IN      A      64.233.170.113
google.com.                281     IN      A      64.233.170.101
google.com.                281     IN      A      64.233.170.100

;; Query time: 2 msec
;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)
;; WHEN: Thu May 14 17:39:25 UTC 2026
;; MSG SIZE rcvd: 135

arunsubramaniyan323@cloudshell:~/nura$ 

```

Chapter 6:

```

arunsubramaniyan323@cloudshell:~/nura$ ps
  PID TTY          TIME CMD
 2134 pts/4    00:00:00 bash
 2853 pts/4    00:00:00 ps
arunsubramaniyan323@cloudshell:~/nura$ 
arunsubramaniyan323@cloudshell:~/nura$ pgrep firefox
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$ 

```

```

arunsubramaniyan323@cloudshell:~/nura$ pkill firefox
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$

arunsubramaniyan323@cloudshell:~/nura$ ps -aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.0  0.0   4324   3264 ?        Ss   16:46   0:00 /bin/bash /google/scripts/onrun.sh sleep infinity
root         9   0.0  0.0  222140   3776 ?        Ssl  16:46   0:00 /usr/sbin/rsyslogd
root        25   0.0  0.1  38768 28124 ?        S    16:46   0:00 /usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --logfile
root        26   0.0  0.0   6196   2708 ?        S    16:46   0:00 logger -t supervisord
root        67   0.0  0.0  12024   4184 ?        Ss+  16:46   0:00 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/keys/authorized_keys [listener] 0 of 10-60 startups
root       301   0.0  0.5 2169572 85344 ?        sl   16:46   0:00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
root       437   0.0  0.0 1234524 12492 ?        sl   16:46   0:00 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.j
root       439   0.0  0.0  16328   6428 ?        S    16:46   0:00 sudo /google/devshell/tmux-agent
root       441   0.0  0.0 1226100 2700 ?        sl   16:46   0:00 /google/devshell/tmux-agent
root       471   0.1  0.2 1867468 46352 ?        Ssl  16:46   0:04 containerd --config /var/run/docker/containerd/containerd.toml
root       677   0.0  0.0   2696   1460 ?        S    16:46   0:00 sleep infinity
root       717   0.0  0.0   9196   4008 ?        S    16:46   0:00 runuser -l arunsubramaniyan323 -c export GOPATH=/home/arunsubramaniyan323/gopath:/google/gopath HOME=/home/arunsubramani
arunsub+   718   0.0  0.0   4324   3276 ?        Ss   16:46   0:00 sh ./bin/codeoss-cloudshell --host=0.0.0.0 --port=971
arunsub+   739   0.2  0.6 11831976 110748 ?        sl   16:46   0:09 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/server-main.j
arunsub+   768   0.0  0.0   7600   3192 ?        Ss+  16:46   0:00 tmux new-session -A -D -n cloudshell -s 631831535
arunsub+   769   0.0  0.0   8800   5932 pts/2    Ss+  16:46   0:00 -bash
arunsub+  1549   0.1  0.4 1127948 73940 ?        sl   16:46   0:04 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-for
arunsub+  1560   0.0  0.0   9020   6056 pts/3    Ss+  16:46   0:00 /bin/bash --init-file /google/devshell/editor/code-oss-for-cloud-shell/out/vs/workbench/contrib/terminal/common/scripts/
root     2119   0.0  0.0  13996   9792 ?        Ss+  17:21   0:00 sshd: arunsubramaniyan323 [priv]
root     2121   0.0  0.0  13996   9720 ?        Ss+  17:21   0:00 sshd: arunsubramaniyan323 [priv]
arunsub+  2123   0.0  0.0  14256   6396 ?        S<   17:21   0:00 sshd: arunsubramaniyan323@pts/0
arunsub+  2124   0.0  0.0   7340   3736 pts/0    Ss+  17:21   0:00 bash -c (sudo touch /var/run/google/devshell/35579 &) trap "sudo rm -f /var/run/google/devshell/35579" EXIT
arunsub+  2127   0.0  0.0   7604   4280 pts/0    Ss+  17:21   0:00 /bin/bash --nrc --noprofile
arunsub+  2128   0.0  0.0  14256   6444 ?        S<   17:21   0:00 sshd: arunsubramaniyan323@pts/1
arunsub+  2131   0.0  0.0   7340   3812 pts/1    Ss+  17:21   0:00 bash -c if [ -f /google/devshell/start-shell.sh ]; then /google/devshell/start-shell.sh '' '' '' '662840578' fals
arunsub+  2132   0.0  0.0   7340   3744 pts/1    Ss+  17:21   0:00 /bin/bash /google/devshell/start-shell.sh 662840578 false
arunsub+  2133   0.0  0.0   6880   3292 pts/1    Ss+  17:21   0:00 tmux new-session -A -D -n cloudshell -s 662840578
arunsub+  2134   0.0  0.0   8800   5920 pts/4    Ss+  17:21   0:00 -bash
arunsub+  2504   2.6  4.3 44461984 719796 ?        sl   17:21   0:34 /google/devshell/editor/code-oss-for-cloud-shell/node --dns-result-order=ipv4first /google/devshell/editor/code-oss-for-
arunsub+  2515   0.0  0.3 1048516 61860 ?        sl   17:21   0:00 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/out/bootstrap-for
arunsub+  2537   0.0  0.3 1016646 62984 ?        sl   17:22   0:00 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/extensions/json-l
arunsub+  2549   0.0  0.4 1943668 72420 ?        sl   17:22   0:00 /home/arunsubramaniyan323/.cache/cloud-code/cloudcode_cli/cloudcode_cli/fba419f4/cloudcode_cli duet -trace -logtostderr
arunsub+  2558   0.5  1.3 32890524 217080 ?        sl   17:22   0:07 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/extensions/gemini
arunsub+  2565   0.6  3.7 1853092 615844 ?        sl   17:22   0:07 /google/devshell/editor/code-oss-for-cloud-shell/node /google/devshell/editor/code-oss-for-cloud-shell/extensions/cloudc
arunsub+  2872   0.0  0.0  10898   4628 pts/4    R<+  17:43   0:00 ps -aux
arunsubramaniyan323@cloudshell:~/nura$

arunsubramaniyan323@cloudshell:~/nura$ bg
-bash: bg: current: no such job
arunsubramaniyan323@cloudshell:~/nura$

arunsubramaniyan323@cloudshell:~/nura$ fg
-bash: fg: current: no such job

arunsubramaniyan323@cloudshell:~/nura$ jobs
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$ nice -n 10 python test.py
false
arunsubramaniyan323@cloudshell:~/nura$
arunsubramaniyan323@cloudshell:~/nura$ renice 5 -p 1234
renice: failed to get priority for 1234 (process ID): No such process
arunsubramaniyan323@cloudshell:~/nura$

```

Chapter 7:

```
arunsubramaniyan323@cloudshell:~/nura$ df
Filesystem                1K-blocks      Used Available Use% Mounted on
overlay                   98831908 76311764   22503760   78% /
tmpfs                     65536          0     65536    0% /dev
/dev/sda1                 98831908 76311764   22503760   78% /root
/dev/disk/by-id/google-home-part1 5018272   153920    4585876    4% /home
/dev/root                 2019696 1288424     731272   64% /usr/lib/modules
shm                       65536          0     65536    0% /dev/shm
arunsubramaniyan323@cloudshell:~/nura$
```

```
arunsubramaniyan323@cloudshell:~/nura$ du -h
12K .
arunsubramaniyan323@cloudshell:~/nura$
```

```
arunsubramaniyan323@cloudshell:~/nura$ sudo mkfs.ext4 /dev/sdb1
mke2fs 1.47.0 (5-Feb-2023)
/dev/sdb1 is apparently in use by the system; will not make a filesystem here!
arunsubramaniyan323@cloudshell:~/nura$
```

```
arunsubramaniyan323@cloudshell:~/nura$ mount
overlay on / type overlay (rw,relatime,lowerdir=206/fs:199/fs:198/fs:196/fs:193/fs:192/fs:190/fs:189/fs:188/fs:187/fs:185/fs:184/fs:183/fs:182/fs:181/fs:180/fs:179/fs:178/fs:177/fs:176/fs:175/fs:174/fs:173/fs:172/fs:163/fs:162/fs:161/fs:160/fs:157/fs:156/fs:155/fs:152/fs:151/fs:148/fs:146/fs:143/fs:142/fs:141/fs:140/fs:139/fs:138/fs:137/fs:136/fs:133/fs:132/fs:128/fs:126/fs:125/fs:123/fs:119/fs:118/fs:116/fs:114/fs:110/fs:107/fs:106/fs:105/fs:104/fs:102/fs:99/fs:97/fs:95/fs:94/fs:91/fs:90/fs:85/fs:84/fs:83/fs:82/fs:80/fs:78/fs:69/fs:68/fs:67/fs:66/fs:65/fs:64/fs:63/fs:62/fs:6/fs,upperdir=/var/lib/containerd/io.containerd.snapshotter.v1.gcfs/snapshotter/318/work)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev type tmpfs (rw,nosuid,size=65536k,mode=755)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=666)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
sysfs on /sys type sysfs (ro,nosuid,nodev,noexec,relatime)
cgroup on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime)
/dev/sda1 on /root type ext4 (rw,relatime,commit=30)
/dev/disk/by-id/google-home-part1 on /home type ext4 (rw,noatime,journal_checksum,errors=remount-ro,data=ordered)
/dev/dm-0 on /usr/lib/modules type ext2 (ro,relatime)
/dev/sda1 on /etc/hosts type ext4 (rw,relatime,commit=30)
/dev/sda1 on /etc/hostname type ext4 (rw,relatime,commit=30)
/dev/sda1 on /etc/resolv.conf type ext4 (rw,relatime,commit=30)
shm on /dev/shm type tmpfs (rw,relatime,size=65536k)
/dev/sda1 on /var/config/tmux type ext4 (ro,relatime,commit=30)
/dev/sda1 on /etc/ssh/keys type ext4 (rw,relatime,commit=30)
/dev/sda1 on /var/lib/docker type ext4 (rw,relatime,commit=30)
/dev/sda1 on /run/google/devshell type ext4 (rw,relatime,commit=30)
none on /sys/kernel/security type securityfs (rw,relatime)
arunsubramaniyan323@cloudshell:~/nura$
```

```
arunsubramaniyan323@cloudshell:~/nura$ sudo umount /dev/sdb1
umount: /dev/sdb1: not mounted.
arunsubramaniyan323@cloudshell:~/nura$
```

```
arunsubramaniyan323@cloudshell:~/nura$ sudo blkid
/dev/sdb1: UUID="b9f0e459-9c03-483b-a0dc-629a47561fb9" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="5245a676-01"
/dev/dm-0: LABEL="ROOT-A" SEC_TYPE="ext2" BLOCK_SIZE="4096" TYPE="ext4"
/dev/sda4: PARTLABEL="KERN-B" PARTUUID="76f23ba4-cdd3-1143-91d2-9a5f7372c111"
/dev/sda2: PARTLABEL="KERN-A" PARTUUID="2762bea5-16a2-5944-b00a-f19370e6ab65"
/dev/sda12: SEC_TYPE="msdos" LABEL_FATBOOT="EFI-SYSTEM" LABEL="EFI-SYSTEM" UUID="7BE5-F6D0" BLOCK_SIZE="512" TYPE="vfat" PARTLABEL="EFI-SYSTEM" PARTUUID="1b9d2508-4204-1a41-81b8-2ddc3bfe9b40"
/dev/sda9: PARTLABEL="reserved" PARTUUID="9634bb25-70d7-1a45-8b09-3648a7de5fe8"
/dev/sda10: PARTLABEL="reserved" PARTUUID="9f5dc8da-5cc5-f44e-8289-f69f69e82b1d"
/dev/sda7: PARTLABEL="ROOT-C" PARTUUID="9b22b5b6-84ac-a94a-b087-3606f5ff48b7"
/dev/sda5: PARTLABEL="ROOT-B" PARTUUID="969395a3-4a21-b54f-a5ae-3a973a048005"
/dev/sda3: LABEL="ROOT-A" SEC_TYPE="ext2" BLOCK_SIZE="4096" TYPE="ext4" PARTLABEL="ROOT-A" PARTUUID="ca1d3ffc-74bb-3a4c-872e-8923b3f79698e"
/dev/sda1: LABEL="STATE" UUID="9f2dbcb8-9534-4465-bea2-481409599ab8" BLOCK_SIZE="4096" TYPE="ext4" PARTLABEL="STATE" PARTUUID="3154d2f0-8b5c-4447-a5e0-0c5002398a44"
/dev/sda11: PARTLABEL="RWFV" PARTUUID="bfed8244-2507-ac48-adfc-b7aa9c3bbbae"
/dev/sda8: LABEL="OEM" UUID="7c475393-34ab-4c9b-aebb-a9cea72447d4" BLOCK_SIZE="4096" TYPE="ext4" PARTLABEL="OEM" PARTUUID="d2d9c06d-32cf-1b49-a5e8-e9c15ebb7dc5"
/dev/sda6: PARTLABEL="KERN-C" PARTUUID="9ea468af-861b-9946-a767-3bfe1912f69e"
arunsubramaniyan323@cloudshell:~/nura$
```

```

arunsubramaniyan323@cloudshell:~/nura$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0  100G  0 disk
├─sda1       8:1    0  95.8G  0 part /run/google/devshell
│           /var/lib/docker
│           /etc/ssh/keys
│           /var/config/tmux
│           /etc/resolv.conf
│           /etc/hostname
│           /etc/hosts
│           /root
├─sda2       8:2    0   16M  0 part
├─sda3       8:3    0    2G  0 part
├─sda4       8:4    0   16M  0 part
├─sda5       8:5    0    2G  0 part
├─sda6       8:6    0  512B  0 part
├─sda7       8:7    0  512B  0 part
├─sda8       8:8    0   16M  0 part
├─sda9       8:9    0  512B  0 part
├─sda10      8:10   0  512B  0 part
├─sda11      8:11   0    8M  0 part
└─sda12      8:12   0   32M  0 part
sdb          8:16    0    5G  0 disk
└─sdb1       8:17   0    5G  0 part /home
arunsubramaniyan323@cloudshell:~/nura$

arunsubramaniyan323@cloudshell:~/nura$ sudo tune2fs -l /dev/sda1
tune2fs 1.47.0 (5-Feb-2023)
Filesystem volume name:   STATE
Last mounted on:         /var
Filesystem UUID:         9f2dbcb8-9534-4465-bea2-481409599ab9
Filesystem magic number:  0xEF53
Filesystem revision #:    1 (dynamic)
Filesystem features:      has_journal ext_attr resize_inode dir_index filetype needs_recovery extent 64bit flex_bg sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Filesystem flags:         signed_directory_hash
Default mount options:    user_xattr acl
Filesystem state:         clean
Errors behavior:          Remount read-only
Filesystem OS type:       Linux
Inode count:              6258720
Block count:              25126395
Reserved block count:     0
Overhead clusters:        418418
Free blocks:              5631974
Free inodes:              4791197
First block:              0
Block size:               4096
Fragment size:            4096
Group descriptor size:    64
Reserved GDT blocks:      609
Blocks per group:         32768
Fragments per group:      32768
Inodes per group:         8160
Inode blocks per group:   510
Flex block group size:    16
Filesystem created:       Fri May  8 10:15:28 2026
Last mount time:          Thu May 14 16:15:05 2026
Last write time:          Thu May 14 17:15:16 2026
Mount count:              6
Maximum mount count:      -1
Last checked:             Thu Nov 19 11:00:00 2009
Check interval:           0 (<none>)
Lifetime writes:          77 GB
Reserved blocks uid:      0 (user root)

Reserved blocks gid:      0 (group root)
First inode:              11
Inode size:               256
Required extra isize:     32
Desired extra isize:      32
Journal inode:            8
Default directory hash:   half_md4
Directory Hash Seed:      239222de-2248-4ca5-8d76-167546212650
Journal backup:           inode blocks
Checksum type:            crc32c
Checksum:                 0x9cd4b250
arunsubramaniyan323@cloudshell:~/nura$

```

Chapter 8:

```

top - 17:57:20 up 1:42, 0 user, load average: 1.32, 0.54, 0.20
Tasks: 37 total, 1 running, 36 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.5 us, 0.3 sy, 0.0 ni, 98.9 id, 0.0 wa, 0.0 hi, 0.2 si, 0.1 st
MiB Mem : 15998.9 total, 12117.4 free, 2212.9 used, 1947.1 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 13785.9 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0    4324    3264    2984  S   0.0   0.0   0:00.08 bash
    9 syslog     20   0   222140   3776    2412  S   0.0   0.0   0:00.08 rsyslogd
   25 root        20   0    38768   28124   11020  S   0.0   0.2   0:01.08 python
   26 root        20   0    6196    2708    2468  S   0.0   0.0   0:00.00 logger
   67 root        10  -10   12024    4184    3072  S   0.0   0.0   0:00.10 sshd
  301 root        20   0   2169572  85348   61532  S   0.0   0.5   0:00.93 dockerd
  437 root        20   0   1234524  12492    8208  S   0.0   0.1   0:01.16 editor-proxy
  439 root        20   0    16328    6428    5504  S   0.0   0.0   0:00.01 sudo
  441 root        20   0   1226100   2700    1988  S   0.0   0.0   0:00.00 tmux-agent
  471 root        20   0   1867468  46352   32532  S   0.0   0.3   0:05.55 containerd
  677 root        20   0    2696    1460    1364  S   0.0   0.0   0:00.00 sleep
  717 root        20   0    9196    4008    3628  S   0.0   0.0   0:00.00 runuser
  718 arunsub+    20   0    4324    3276    3012  S   0.0   0.0   0:00.00 sh
  739 arunsub+    20   0   11.3g  110872   51192  S   0.0   0.7   0:09.72 node
  768 arunsub+   10  -10    8076    3584    2084  S   0.0   0.0   0:00.22 tmux
  769 arunsub+   10  -10    8800    5932    4060  S   0.0   0.0   0:00.07 bash
 1549 arunsub+   20   0   1127948  73940   48320  S   0.0   0.5   0:05.23 node
 1560 arunsub+   20   0    9020    6056    4028  S   0.0   0.0   0:00.04 bash
 2119 root        10  -10   13996    9792    8416  S   0.0   0.1   0:00.01 sshd
 2121 root        10  -10   13996    9720    8340  S   0.0   0.1   0:00.01 sshd
 2123 arunsub+   10  -10   14256    6396    4740  S   0.0   0.0   0:00.00 sshd
 2124 arunsub+   10  -10    7340    3736    3484  S   0.0   0.0   0:00.00 bash
 2127 arunsub+   10  -10    7604    4280    3664  S   0.0   0.0   0:00.00 bash
 2128 arunsub+   10  -10   14256    6444    4784  S   0.0   0.0   0:00.14 sshd
 2131 arunsub+   10  -10    7340    3812    3560  S   0.0   0.0   0:00.00 bash
 2132 arunsub+   10  -10    7340    3744    3480  S   0.0   0.0   0:00.00 start-shell.sh
 2133 arunsub+   10  -10    6880    3292    2992  S   0.0   0.0   0:00.00 tmux
 2134 arunsub+   10  -10    8932    6016    4040  S   0.0   0.0   0:00.20 bash
 2504 arunsub+   20   0   42.4g  719888   62332  S   0.0   4.4   0:35.24 node
 2515 arunsub+   20   0   1048516  61936   47988  S   0.0   0.4   0:00.38 node
 2537 arunsub+   20   0   1016648  63088   46440  S   0.0   0.4   0:00.35 node

0% Tasks: 37, 108 tbr, 0 kthr, 1 running
1% Load average: 1.40 0.62 0.23
2% Uptime: 01:42:43
3%
Mem[|||||] 1.89G/15.6G
Swp[|||||] 0K/0K

Name: 5/G
  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM    TIME+  Command
 3001 arunsub+   10  -10    8632   4872   3600  S  200.0   0.0   0:00.03 httpd
    1 root        20   0    4324    3264    2984  S   0.0   0.0   0:00.06 /bin/bash /google/scripts/onrun.sh sleep infinity
    9 syslog     20   0    216M    3776    2412  S   0.0   0.0   0:00.00 /usr/sbin/rsyslogd
   10 syslog     20   0    216M    3776    2412  S   0.0   0.0   0:00.03 /usr/sbin/rsyslogd
   11 syslog     20   0    216M    3776    2412  S   0.0   0.0   0:00.00 /usr/sbin/rsyslogd
   12 syslog     20   0    216M    3776    2412  S   0.0   0.0   0:00.03 /usr/sbin/rsyslogd
   25 root        20   0   38768   28124   11020  S   0.0   0.2   0:01.08 /usr/bin/python /usr/bin/supervisord -n -c /google/devshell/supervisord.conf --pidfile=/var/run/supervisor.pid --logfile
   26 root        20   0    6196    2708    2468  S   0.0   0.0   0:00.00 logger -t supervisord
   67 root        10  -10   12024    4184    3072  S   0.0   0.0   0:00.10 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/keys/authorized_keys [listener] 1 of 10-60 startups
  301 root        20   0   2118M   85348   61532  S   0.0   0.5   0:00.19 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  408 root        20   0   2118M   85348   61532  S   0.0   0.5   0:00.13 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  409 root        20   0   2118M   85348   61532  S   0.0   0.5   0:00.01 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  410 root        20   0   2118M   85348   61532  S   0.0   0.5   0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  411 root        20   0   2118M   85348   61532  S   0.0   0.5   0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  412 root        20   0   2118M   85348   61532  S   0.0   0.5   0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  436 root        20   0   2118M   85348   61532  S   0.0   0.5   0:00.07 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
  437 root        20   0   1205M   12492    8208  S   0.0   0.1   0:00.16 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-c
  450 root        20   0   1205M   12492    8208  S   0.0   0.1   0:00.00 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-c
  451 root        20   0   1205M   12492    8208  S   0.0   0.1   0:00.00 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-c
  471 root        20   0   1823M   46352   32532  S   0.0   0.3   0:00.05 containerd --config /var/run/docker/containerd/containerd.toml=/google/devshell/editor/code-oss-for-cloud-shell/static-c
  472 root        20   0   2118M   85348   61532  S   0.0   0.5   0:00.15 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registry-mirror=https://asia-mirror.gcr.io-for-cloud-shell/static-c
  474 root        20   0   1823M   46352   32532  S   0.0   0.3   0:01.64 containerd --config /var/run/docker/containerd/containerd.toml=/google/devshell/editor/code-oss-for-cloud-shell/static-c
  475 root        20   0   1823M   46352   32532  S   0.0   0.3   0:00.96 containerd --config /var/run/docker/containerd/containerd.toml=/google/devshell/editor/code-oss-for-cloud-shell/static-c
  476 root        20   0   1823M   46352   32532  S   0.0   0.3   0:00.93 containerd --config /var/run/docker/containerd/containerd.toml
  477 root        20   0   1823M   46352   32532  S   0.0   0.3   0:00.00 containerd --config /var/run/docker/containerd/containerd.toml
  478 root        20   0   1823M   46352   32532  S   0.0   0.3   0:00.00 containerd --config /var/run/docker/containerd/containerd.toml
  492 root        20   0   1823M   46352   32532  S   0.0   0.3   0:00.36 containerd --config /var/run/docker/containerd/containerd.toml

procs -----memory----- --swap-- -----io---- -system-- -----cpu-----
 r  b  swpd  free  buff  cache  si  so    bi  bo    in  cs  us  sy  id  wa  st  gu
  1  0      0 12414436 87448 1908200  0  0    287  51 1056  4  1  1 98  0  0  0
arunsubramaniyan323@cloudshell:~/nura$
arunsubramaniyan323@cloudshell:~/nura$ free
              total            used            free              shared        buff/cache          available
Mem:          16382836          2263816          12408900              1148            1995780            14119020
Swap:              0              0              0
arunsubramaniyan323@cloudshell:~/nura$
arunsubramaniyan323@cloudshell:~/nura$ uptime
 18:00:13 up 1:45, 0 user, load average: 1.35, 0.89, 0.39
arunsubramaniyan323@cloudshell:~/nura$

```

Chapter 9:

```

arunsubramaniyan323@cloudshell:~/nura$ sudo useradd arun
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$ █

--
arunsubramaniyan323@cloudshell:~/nura$ sudo userdel arun
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$ █

arunsubramaniyan323@cloudshell:~/nura$ groups
arunsubramaniyan323 adm sudo docker
arunsubramaniyan323@cloudshell:~/nura$ █

-
arunsubramaniyan323@cloudshell:~/nura$ sudo groupadd developers
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$ █

--
arunsubramaniyan323@cloudshell:~/nura$ sudo groupdel developers
arunsubramaniyan323@cloudshell:~/nura$ ls
demo.py  main  test.py
arunsubramaniyan323@cloudshell:~/nura$ █

```

Chapter 10:

```

arunsubramaniyan323@cloudshell:~/nura$ bzip2 test.py
bzip2: Input file test.py has 1 other link.
arunsubramaniyan323@cloudshell:~/nura$ █

arunsubramaniyan323@cloudshell:~/nura$ bunzip2 file.txt.bz2
bunzip2: Can't open input file file.txt.bz2: No such file or directory.
arunsubramaniyan323@cloudshell:~/nura$ █

arunsubramaniyan323@cloudshell:~/nura$ xz test.py
xz: test.py: Input file has more than one hard link, skipping
arunsubramaniyan323@cloudshell:~/nura$ █

--
arunsubramaniyan323@cloudshell:~/nura$ tar -cvf archive.tar folder/
tar: folder: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
arunsubramaniyan323@cloudshell:~/nura$ █

arunsubramaniyan323@cloudshell:~/nura$ tar -xvf archive.tar
arunsubramaniyan323@cloudshell:~/nura$ tar -czvf archive.tar.gz folder/
tar: folder: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
arunsubramaniyan323@cloudshell:~/nura$ tar -xzvf archive.tar.gz
arunsubramaniyan323@cloudshell:~/nura$ █

```

```
arunsubramaniyan323@cloudshell:~/nura$ tar -xzvf archive.tar.gz
arunsubramaniyan323@cloudshell:~/nura$ zip archive.zip file.txt
zip warning: name not matched: file.txt

zip error: Nothing to do! (archive.zip)
arunsubramaniyan323@cloudshell:~/nura$ 

arunsubramaniyan323@cloudshell:~/nura$ unzip archive.zip
unzip: cannot find or open archive.zip, archive.zip.zip or archive.zip.ZIP.
arunsubramaniyan323@cloudshell:~/nura$ 

arunsubramaniyan323@cloudshell:~/nura$ lzma test.py
lzma: test.py: Input file has more than one hard link, skipping
arunsubramaniyan323@cloudshell:~/nura$ 
```